TECHNICAL SPECIFICATION OF TESTING INSTRUMENTS

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I. FUSION SPLICING MACHINE

Fusion splicer shall be of state of the art single mode and multimode which can complete a splice sequence together with detailed loss estimation in less than 15 seconds. Fusion splicer shall have the following Features:

1. Core-to core Fiber alignment using profile alignment system technology
2. Environmental resistance features
3. most compact and lightest
4. Arc calibration-free system
5. 4.1'' TFT Color LCD monitor
6. Dual directional operation system
7. Long life Battery
8. Fibre holder option
9. Software upgrade via internet
10. Detachable work table

Specification :

i.	Applicable Fiber	:	SM / MM/DS and other single mode fibres
ii.	Fiber Count	:	Single
iii.	Fiber cleaved length		
	Standard	:	16 mm
	Fiber holder	:	FH-60-250 and FH-60-900 (10mm cleave)
iv.	Average Fusion loss	:	0.01 d B (SM), 0.01 d B (MM), 0.04 d B (DS)
v.	Storage of splice result	:	2000 or more splice results
	Working mode	:	Auto & Manual
vi.	Splicing time	:	< 15 sec.
vii.	Return loss	:	> 60 d B
viii.	No. of splice modes	:	100 modes.
ix.	Splice loss estimating function	:	should be available.
	Attenuation splicing function	:	Auto attenuation mode :0.1 d B to 15 d B (0.1 dB step)
x.	Tension test	:	1.96 to 2.25N
xi.	Fiber Magnification	:	132 (X/Y Simultaneous view) or 264 (high magnification view)
xii.	Operational environmental condition	:	0 to 5000 m above sea level, 0 to 95% RH and10 to 50°C and upto 15m/s wind velocity.
xiii.	Storage condition	:	0 to 95% RH and –40 to 80°C respectively
xiv.	Tube heater	:	Built-in tube heater with 30 programmable heating modes.
xv.	Applicable protection sleeve length	:	60 mm, 40 mm and micro sleeves
xvi.	No.of splice cycles with battery	:	Typical 160 cycles
xvii.	Electrode life	:	2500 arc discharges.

xviii. Viewing method and display : Two CMOS cameras and 4.1'' TFT Color LCD monitor

xix. Power supply : 100 to 240 V AC, with Battery pack; shall operate on battery at least for 4 hours.

Standard Accessories (These accessories should be supplied in FSM package)

1. AC adapter / Battery charger
2. AC power cord
3. DC Power cord
4. Spare electrodes
5. USB cable
6. user manual
7. Battery pack
8. Battery charge cord
9. High precision fibre cleaver
10. J plate
11. Carrying case

GTP FOR FUSION SPLICING MACHINE

S.No.	Description	
1.	Make & Model	:
2.	Fibre Count	:
3.	Applicable Fibres	:
4.	Average Fusion Loss	:
5.	Storage of Splice Result	:
6.	Working mode	:
7.	Splicing time	:
8.	Return Loss	:
9.	No. of Splice modes	:
10.	Splice loss estimating function	:
11.	Attenuation splicing function	:
12.	Fibre magnification	:
13.	Tension test	:
14.	Tube Heater (heater modes)	:
15.	Power Supply	:
16.	Battery Backup capacity	:
17.	Carrying case	:
18.	Dimensions	:
19.	Weight	:
20.	Operating condition	:
21.	Storage condition	:
22.	No. Of splice cycles with battery	:
23.	Electrode life	:
24.	Accessories	:

Signature of the bidder.

II. **OPTICAL TIME DOMAIN** REFLECTOMETER (OTDR)

To install & maintain optical fiber system and for auto fiber analysis the Optical Time Domain Reflector (OTDR) is required. The OTDR offered should be rugged and easy to use with intelli-trace facility. It should be designed to stand the rigors of field work, while providing precise measurements on single mode or multimode optical fiber systems.

TECHNICAL CHARACTERISTICS:

Optical characteristics	:	1550 nm Optical Output 1550 +/- 20 nm
		Loss threshold 0.02 dB minimum
		Loss Resolution : 0.001 d B
Dynamic and Measurement range	:	Dynamic Range Measurement Range
1310 /1550 NM		45/45 dB
Readout Resolution (Horizontal)	:	0.1m Minimum
Distance measurements	:	Kilometres, Feet, Miles
OTDR Distance Range settings	:	Single mode – 1 to 240 km
		Multi mode - 1 to 40 km
System Measurement accuracy	:	Distance - +/- 4 m
		Loss : +/- 0.01 dB
Distance Resolution	:	25 cm to 40 m
Measurement points (horizontal)	:	up to 256000 sampling points
Vertical Scale	:	0.1 to 10.0 dB/Div
Read out resolution (Vertical)	:	0.001 dB
Reflectance range (Vertical)	:	-14dB to – 60dB

Distance accuracy	:	+/- 1.25m
Reflectance range	:	-14dB to – 60 dB
Measurement Time	:	Less than 3 minutes
(Intelli Trace Technology)	:	(20 d B accumulated loss)

Dead Zone for 1310 & 1550 nm

	Event Dead zone	Attenuation dead zone
Single Mode	3 m	10/12 m

Display	:	Color screen VGA-LCD
Selectable pulse widths	:	10ns, 30ns, 100ns, 300ns, 1µs, 3µs and 10 µs
Memory capacity	:	Internal Memory:1GB WiFi/Bluetooth : Standard IEEE 802.11b/g And blue tooth class 2
Output ports	:	Ethernet: 10/100/1000Mhz USB: 2*usb 2.0ports
Power	:	AC 100-240 Vrms ± 10%, 50-60Hz. DC Operation Battery Operation : Should be provided with rechargeable Battery pack (5 hours of operation typical, depending on operating mode)
Operating temperature	:	0°C to 50°C
Storage Temperature	:	-40°C to +60°C
Accessories	:	AC/DC Adapter, Power cord, NimH Battery pack, User's manual, RS232 cable, Shoulder strap, Carrying case

GTP PARTICULARS FOR OTDR

S.NO.	Description	
1.	Optical characteristics	:
2.	Dynamic and Measurement range 1310 /1550 NM	:
3.	Readout Resolution(Horizontal)	:
4.	Distance measurements	:
5.	OTDR Distance Range settings	:
6.	System Measurement accuracy	:
7.	Distance Resolution	:
8.	Measurement points (horizontal)	:
9.	Vertical Scale	:
10.	Read out resolution (Vertical)	:
11.	Reflectance range (Vertical)	:
12.	Distance accuracy	:
13.	Reflectance range	:
14.	Measurement Time	:
15.	Dead Zone for 1310 & 1550 nm Single Mode	Event Dead zone Attenuation dead zone
16.	Display	:

17. Selectable pulse widths

18. Memory capacity :

19. Output ports :

20. Power :

21. Operating temperature :

22. Storage Temperature :

23. Accessories Whether included (YES/NO) :

Signature of the Bidder

III. OPTICAL POWER METER

The Optical Power Meter offered should be operatable in windows used in Telecommunication system and shall be suitable for measuring the power of Optical signal sources and the attenuation of optical cables. The equipment shall be suitable for installation and maintenance of optical links and for component characterization.

Technical Specification:

Wavelength range	:	850 to 1550 nm
Accuracy	:	± 5% ± 1 nW
Measurement Range	:	-70 to +10 dBm
Operating Temperature	:	- 10 to + 50° C
Storage Temperature	:	- 20 to + 70° C
Optical Connector Type	:	FC
Operating time with battery	:	At least 10 hours

Optical Power Meter should be operated on DC voltage. Suitable AC/DC Adapter and battery shall be provided.

Accessories along with power meter

1. FC/PC, SC/PC, ST/PC interchangeable connectors
2. Battery pack
3. Carrying Bag
4. AC/DC adapter.
5. User Guide

GTP FOR OPTICAL POWER METER

S.NO.	Description	
1.	Wavelength range	:
2.	Accuracy at - 30 dBm 23 + 1deg. C,	:
	1550nm	:
3.	Measurement Range	:
4.	Operating Temperature	:
5.	Storage Temperature	:
6.	Optical Connector Type	:
7.	Operating time with battery	:
8.	Weight	:
9.	Dimensions	:

Signature of the bidder with seal

IV. OPTICAL LASER SOURCE

The Optical Laser Source should transmit optical signal with specific selected wavelength and power level into a fiber optic link. Together with the optical power meter, it should measure the optical loss or attenuation of the link.

The Optical Laser Source should have Automatic wavelength switching, which allows the unit to automatically toggle between available wavelengths. Using this source with a power meter, it should recognize the wavelength in use and switch to the proper calibration parameter.

The Optical Laser Source should be Accurate and Dual Mode, which offer excellent stability and portability for accurate fiber optic testing with Single output port to provide stable laser power at dual wavelengths. It can operate in either continuous wave (CW) mode or modulated mode. A low battery indicator shall remind the user to recharge the battery.

Technical Specification:

Output wavelength (nm)	: 1310&1550
Central Wavelength	: 1310±20nm & 1550±20nm
Power Stability	:
Short Term (15 min.)	: < ±0.03dB @ 1310 & 1550nm
Long Term (8 Hours)	: < ±0.1 dB @ 1310 &1550nm
Spectral Width	: ≤5nm
Output frequency (Hz)	: 270,1K, 2K
Output Power (dBm)	: ≥1
Auto Power-off	: Yes
Back-light	: Yes
Operating Temperature	: -10 to+50oC
Storage Temperature	: -40 to+70oC
Operating time with battery	: At least 50 hours in Auto mode
Relative humidity	: 0% to 95% non-condensing

The Optical Laser Source should be operated on DC voltage. Suitable AC/DC Adapter and battery shall be provided.

Accessories to be provided along with Optical Laser Source

1. FC/PC, SC/PC, ST/PC interchangeable connectors
2. Battery pack
3. Carrying Bag
4. AC/DC adapter.
5. User Guide

GTP FOR OPTICAL LASER SOURCE

S.NO.	Description	
1.	Output wavelength (nm)	:
2.	Central Wavelength	:
3.	Power Stability	:
	Short Term (15 min.)	:
	Long Term (8 Hours)	:
4.	Spectral Width	:
5.	Output frequency (Hz)	:
6.	Output Power (dBm)	:
7.	Auto Power-off	:
8.	Back-light	:
9.	Operating Temperature	:
10.	Storage Temperature	:
11.	Operating time with battery	:
12.	Relative humidity	:
13.	Optical Connector Type	:
14.	Weight	:
15.	Dimensions	:

Signature of the bidder with seal

V. 4 $^{1/2}$ Digit Handheld Digital Multimeter

1.	Scale	:	4 $^{1/2}$ Digit
2	Count display	:	19999 counts LCD display
3.	DC voltage accuracy	:	± (0.05% +1)
4.	AC voltage accuracy	:	± (0.8% +2)
5.	Ohm accuracy	:	± (0.3% +1)
6.	Frequency response	:	100 kHz
7.	Duty Cycle accuracy	:	± (0.2% per Khz+0.1%)
8.	Ranges		
	i. DC voltage	:	80 mV to 1000V
	ii. AC voltage	:	800 mV to 1000V True (RMS)
	iii. AC & DC current	:	80 mA to 10A
	iv. Resistance	:	800 Ohms to 50M Ohms
	v. Capacitance	:	1nF to 9999 μF
	vi. Frequency	:	50 Hz to 200 KHz
9.	Temperature measurement	:	-200 ^{0}C – 1000 ^{0}C -300 ^{0}F - 2000 ^{0}F Excluding probe
10.	Conductance	:	Maximum conductance 60.00 nS Accuracy ± (1.0% +10)
11.	Diode	:	Range 3V Accuracy ± (2%+1)
12.	Operating temperature	:	-20 ^{0}C to + 55 ^{0}C
13.	Storage temperature	:	- 40 ^{0}C to + 60 ^{0}C
14.	Operating Voltage	:	On a built-in 9V rechargeable battery

15. The digital multimeter to be supplied under the above specification must have overload protection and should be compact in construction and light weight and rugged shock absorbent case.

16. The digital multimeter should have the following indications.

- Automatic Polarity
- Low battery indication.

17. The 4 $^{1/2}$ digit multimeter should be suitable for carrier signal measurement in dBs on PLC, VHF and UHF Communication system and should be capable of diode testing.

18. Accessories along with each meter.

Fuses	2 of each type
Testing probes	2 pairs
User manual	1
Batteries installed	1set

GUARANTEED TECHNICAL PARTICULARS

FOR 4 $^{1/2}$ DIGIT HANDHELD MULTIMETER

(TO BE FILLED BY BIDDER)

Sl.No.	Description	Data
1.	Manufacturer Name & Address	
2.	Suppliers Name & Address	
3.	Make & Type	
4.	Scale	
5.	Count display	
6.	DC voltage accuracy	
7.	AC voltage accuracy	
8.	Ohm accuracy	
9.	Frequency response	
10.	Ranges	
	i. DC voltage :	
	ii. AC voltage :	
	iii. AC & DC current :	
	iv. Resistance :	
	v. Capacitance :	
11.	Weight	
12.	Dimensions	
13.	Whether over load protection provided	Yes/No
14.	Whether capable of indicating automatic Polarity and Low battery	Yes/No
15.	Whether accessories provided	Yes/No

Signature of the bidder.

VI. Digital Earth Resistance Tester

Earth Resistance Range	20Ω, 200 Ω , 2000 Ω
Measuring range	100mΩ to 2000 Ω in 2 ranges
Accuracy	± 1.5% of reading ±5 digits valid from 10% of reading to 95% of the range
Test Frequency	128Hz ± 0.5Hz (Approx)
Test Current	For 20 Ω Range – 10mA AC rms, 200 Ω Range – 1 mA AC rms, 2000 Ω Range – 100µA AC rms
Interference	Interference voltages of 20V±5% peak-to-peak, 50Hz in the potential circuit will have a maximum effect of ±1% on the reading obtained for the 20 Ω to 2k Ω ranges
Earth Voltage Range	0-200V AC, Resolution 0.1V, Accuracy 2% of range
Maximum Output Voltage	Less than 40Vrms
Voltage withstand	In the event of a system fault, the instrument will withstand 240V AC, applied between any two terminals
Operating Temperature	0° C to 55°C Less than 90% RH
Storage Temperature	-20° C to 70° C Less than 90% RH
Influence of Temperature	<± 0.2% per °C over the temperature change0-20°C and 28-55°C
Humidity Operating Storage	 95% RH non condensing max at 40 °C 93% RH max at 55 °C
Dielectric Strength	2KV AC applied between all the terminals and the body of the instrument for one minute°
Display	3 ½ digits with LCD display max Reading 1999
Power	Rechargeable Battery Pack 1.2V, 600mAh Battery (Qty 8 No.) with Battery Charger with Over Charge Protection
Charging Time	Approximate 16 Hrs
Single Charger Battery life	Approximate 5-6 Hrs of continuous operation on mid ranges
Standard Accessories	1) Operating Instruction manual 2) Battery Charger 3) Carrying bag containing a) Four Spikes of 10 mm dia, 450 mm long, b) 10m,20m,30m & 40m of cable on a winder as standard c) Hammer

GUARANTEED TECHNICAL PARTICULARS

DIGITAL EARTH RESISTANCE TESTER

(TO BE FILLED BY BIDDER)

Earth Resistance Range	
Measuring range	
Accuracy	
Test Frequency	
Test Current	
Interference	
Earth Voltage Range	
Maximum Output Voltage	
Voltage withstand	
Operating Temperature	
Storage Temperature	
Influence of Temperature	
Humidity Operating Storage	
Dielectric Strength	
Display	
Dimension	
Power	
Charging Time	
Single Charger Battery life	
Weight	
Standard Accessories	

Signature of the bidder.

VII. OFC Tool kit with Binoculars

TOOLS FOR CUTTING, STRIPPING, SHEATHING, JACKET ARMOURING STRENGTH MEMBERS OF OFC

Each set should consist of the following tools housed in a standard quality suitcase:

1. Fiber Cleaver
2. Fibre Stripper
3. Buffer Tube Stripper
4. Universal Continuity Tester
5. Round Cable Slitter
6. Round Cable Cutter (3-28 mm size)
7. Kevlar Scissors
8. Connectors Cleaner-Cletop
9. Foam Swabs
10. Matching Gel
11. KM Wipes
12. Utility Knife
13. Tweezers
14. Needle Nose Plier
15. Plano Wire
16. 4 bit Screw Driver
17. Black Marker
18. Safety Glasses
19. Black Work Mat
20. Ruler
21. ½" Nut Driver
22. Fusion splice Sleeves (100 No.)
23. PVC Electrical Tape
24. Alcohol Bottle
25. Rugged Carry Case
26. Safety Belts
27. Fiber optic wipes
28. Cleaning Solvent
29. Jacket Stripper
30. 6" Side Cutting plier
31. Cleaning Tissue
32. Degreaser Wipes
33. Fabric Tape Measure
34. Fibre disposal unit
35. Economy tie labels
36. Shrink tube kit
37. 1 Mtr 3mm Furcation
38. 1 Mtr 900 um Furcation
39. Wire Marker dispenser
40. Binoculars

I. HIGH PRECISION CLEAVER:

High precision cleaver should have flexible design which permits it to meet variety of cleaving requirements. This cleaver will be used for fusion splicing. Hence the cleaving mechanism should give a splice loss less than 0.02 d B. It should also have the provision to accept fiber holders to support multi-fiber cleaving operations.

II. STRIPPER:

Stripper shall be designed in such a way that all sizes of fibers to remove without scratching or marring the fiber. It is of hardened, precision formed, ground cutting surfaces and having a 0.005" precision laser drilled hole.

III. BINOCULARS

Binoculars are required for Monitoring of EHT line Towers and its members and position of Fibre Optic Cable and its accessories fixed to the Tower Members.

Binoculars should able to be provide clear picture / image over a minimum distance of 1000 mts. It should also meet the following parameters.

i)	Magnification	:	10 x or better
ii)	Diameter of the front lens	:	50 MM or more

iii) Complete water proof protection
iv) Dry Nitrogen – purged and sealed to prevent fogging, clouding and moisture damage.
v) Fully multi – coated optics which maximizes light transmission for superior brightness and clarity.
vi) BAK-4 prisms for crisp and clear images.
vii) Extreme close focus distance.
viii) Wide field of view
ix) Optimum long eye relief with twist-up eye-cups.
x) Large Knurled center focus knob for precise focusing.
xi) Textured and rugged shock-absorbing rubber armor
xii) Non-slip and ergonomic rubber design for a secure grip.
xiii) Solid and sturdy construction.
xiv) Deluxe carrying case and neck strap.

GTP FOR BINOCULARS

S.No.	Description	
1.	Make & Model	:
2.	Magnification	:
3.	Diameter of the Front lens	:
4.	Water proof protection	:
5.	Focus distance	:
6.	Clear view Distance(kms)	:
7.	Maximum Viewing distance.	:
8.	Shock absorbing Mechanism	:
9.	Prisms design	:
10.	Carrying case	:

Signature of the bidder
with official seal

VIII. Medium Size Tool kit of Taparia make

TOOLS FOR CUTTING, STRIPPING, SHEATHING, JACKET ARMOURING STRENGTH MEMBERS OF WIRES

Each set should consist of the following tools housed in a standard quality suitcase:

1. Screw Driver with detachable bits
2. Spanner set
3. Cutting plier
4. Noseplier-6 inch
5. Wire stripper and cutter: Stripper and cutter shall be designed in such a way that to remove insulation and cutting all sizes of wires/copper cables
6. 25 Watt Soldering Iron
7. 50 Watt Soldering Iron
8. Crimping Tool 4/6/8 pin suitable for 2.5/4/16/25 sq.mm copper cable and RJ45, RJ11 & RJ12 connectors
9. $3^{1/2}$ Digit Multi meter of reputed make company.